Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	7 July 2026
Team ID	**SWTID-2026-2886**
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Import	Import electricity consumption dataset from MySQL into Tableau. Validate data connection before visualization.
FR-2	Data Preparation	Clean and organize electricity data. Convert date fields and remove missing or duplicate values.
FR-3	Interactive Dashboard	Display state-wise, region-wise and yearly electricity consumption. Provide interactive maps, bar charts and pie charts.
FR-4	Data Filters & Analysis	Apply State, Region, Year and Date filters. Support Top N / Bottom N analysis and lockdown comparison using Tab
FR-5	Story & Publishing	Create Tableau Story for insights. Publish Dashboard and Story on Tableau Public and embed into Flask web application.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The dashboard should be simple, interactive,and easy for user knowledge.
NFR-2	**Security**	Only authorized users should be able to modify the project files, while published dashboards remain view-only for the public.

NFR-3	**Reliability**	The dashboard should display accurate electricity consumptio analytical results.
NFR-4	**Performance**	Dashboard filters, parameters, and charts should respond quickly and load efficiently even with large datasets.
NFR-5	**Availability**	The published Tableau dashboard should be accessible online through Tableau Public and the Flask web application whenever required.
NFR-6	**Scalability**	The solution should support additional years, states, regions, and future electricity datasets.